

A novel lattice Boltzmann model for the Poisson equation

Zhenhua Chai ^a, Baochang Shi ^{b,*}

^a State Key Laboratory of Coal Combustion, Huazhong University of Science and Technology, Wuhan 430074, PR China

^b Department of Mathematics, Huazhong University of Science and Technology, Wuhan 430074, PR China

Received 1 May 2007; received in revised form 1 June 2007; accepted 20 June 2007

Available online 10 July 2007

Abstract

In this paper, a novel lattice Boltzmann model is proposed to solve the Poisson equation through modifying equilibrium distribution function. Compared with previous models, which can be viewed as the solvers to diffusion equation, the present model is a genuine solver to the Poisson equation, and the transient term derived by previous models is eliminated. Numerical solutions agree well with analytical solutions, which indicates the potential of the present model for solving the Poisson equation.

© 2007 Elsevier Inc. All rights reserved.

Keywords: Lattice Boltzmann method; The Poisson equation; Chapman–Enskog expansion

1. Introduction

The Poisson equation is a typical partial differential equation (PDE) of elliptic type with a unified form: $\Delta u = R$, where Δ is the Laplace operator, u is the physical quantity, and R is the source term, which is a function of u or position. It deals with the physical potential such as thermodynamic or electrostatic one and describes many potential related phenomena. With the growing interest in bio-MEMS and bio-NEMS applications and fuel cell technologies, electrokinetic flow has become one of the most important nonmechanical techniques in micro-and nanofluidics. In these fields, besides conventional Navier–Stokes equations for depicting fluid flow, there is another equation, i.e., the Poisson–Boltzmann equation, for describing distribution of electrical potential [1–4]. How to solve the Poisson equation efficiently and accurately has been an important issue in both basic and applied science. In the past several years, many numerical methods have been developed for this problem, such as finite difference method (FDM) [5,6], finite element method (FEM) [7,8], boundary element method (BEM) [9] and so on.

As a mesoscopic numerical approach, the lattice Boltzmann method (LBM) has attained increasing attention in simulating complex fluid flows [10–12]. Historically, the LBM originates from lattice gas automaton (LGA) [13], and later it was shown that LBM can be derived from continuous Boltzmann equation [14,15]. Unlike traditional numerical methods, such as the FDM, FEM and BEM, which directly solve the partial

* Corresponding author.

E-mail address: sbchust@126.com (B. Shi).

differential equations (PDE) to compute the desired physical quantities, these required physical quantities are derived with a sum of the distribution function in LBM. In addition, the corresponding PDE can be recovered correctly through the Chapman–Enskog expansion.

Recently, LBM has been extended successfully to solve advection–diffusion equation [16,17], reaction–diffusion equation [18], some nonlinear evolution equations [19–21], wave equation [22] and the Poisson equation [23], which indicates that LBM is not limited to solve hydrodynamic equations.

To the best knowledge of authors, almost all previous lattice Boltzmann models [1–3,23] used to solve the Poisson equation have following common characters: (1) The final equation derived from these models is diffusion equation, not the Poisson equation. (2) Since the diffusion equations are recovered from these models, some unexpected errors may be induced by initialization when these models are used to solve the Poisson equation.

Based on cellular automaton Poisson solver, Hirabayashi et al. [23] proposed a lattice BGK model to solve the Poisson equation. However, this model was a genuine solver to diffusion equation, and what is more, the source term was linear or a fluctuation near zero. The model was hardly used directly to solve nonlinear Poisson–Boltzmann equation [3]. Recently, several lattice Boltzmann models [1–3] were proposed to solve the Poisson–Boltzmann equation, but the key idea among them was same, i.e., the diffusion solver was used to solve the Poisson equation. Although the problem of source term in Hirabayashi’s model was solved, these models were also the solvers to diffusion equation, and the transient term $\frac{\partial u}{\partial t}$ was not eliminated. In the present work, the models mentioned above are also called as artificial models. The objective of this study is to propose a novel and general model to overcome these shortcomings.

The rest of the paper is organized as follows. In Section 2, a novel lattice Boltzmann model for the Poisson equation is introduced. In Section 3, numerical experiments are performed to test the capacity of the present model, and some differences with previous models are also discussed. Summary and conclusion are presented in Section 4.

2. A novel lattice Boltzmann model

A novel and unified lattice Boltzmann model is proposed to solve d -dimension Poisson equation. Considering a generalized case, $DdQq$ model, where d, q are numbers of dimension and possible velocity, respectively, the evolution equation of the density distribution function in LBM reads

$$f_i(x + c_i \Delta t, t + \Delta t) - f_i(x, t) = \Omega_i + \Omega'_i, \quad (2.1)$$

where $\Omega_i = -1/\tau [f_i(x, t) - f_i^{(eq)}(x, t)]$, τ is dimensionless relaxation time, $\Omega'_i = \Delta t \bar{\omega}_i R D$, where R is the right hand side term in the Poisson equation and equals zero in the Laplace equation, $c_i = c e_i$, where e_i is the possible velocity direction in space, $c = \Delta x / \Delta t$, Δx and Δt are the lattice spacing and time step, respectively. D is artificial diffusion coefficient, which is related to dimensionless relaxation time τ , the relation between them can be expressed as $D = \alpha c^2 (\frac{1}{2} - \tau) \Delta t$, where α is a constant which depends on the model used in numerical simulations and can be derived from the following Eq. (2.3). $f_i^{(eq)}(x, t)$ is the equilibrium distribution function, and defined by

$$f_i^{(eq)}(x, t) = \begin{cases} (\omega_0 - 1.0)u(x, t), & i = 0, \\ \omega_i u(x, t), & i = 1 - q, \end{cases} \quad (2.2)$$

where ω_i are weight coefficients, and there are some constraints on ω_i and $\bar{\omega}_i$,

$$\begin{aligned} \sum_{i=0}^q \omega_i &= 1, \\ \sum_{i=0}^q \omega_i c_i &= 0, \\ \sum_{i=0}^q \omega_i c_i c_i &= \alpha c^2 I, \\ \sum_{i=0}^q \bar{\omega}_i &= 1, \end{aligned} \quad (2.3)$$

where I is the unit matrix. With the aids of Eqs. (2.2) and (2.3), we have

$$\sum_{i=0}^q f_i(x, t) = \sum_{i=0}^q f_i^{(\text{eq})}(x, t) = 0. \quad (2.4)$$

Following the method proposed in Ref. [24], the physical quantity u is defined by

$$u(x, t) = \frac{1}{1 - \omega_0} \sum_{i=1}^q f_i^{(\text{eq})}(x, t) = \frac{1}{1 - \omega_0} \sum_{i=1}^q f_i(x, t) \quad (2.5)$$

and the process for recovering the Poisson equation can be summarized as follows. By using the Chapman–Enskog expansion [25] in time and space, we have

$$f_i = f_i^{(\text{eq})} + \epsilon f_i^{(1)} + \epsilon^2 f_i^{(2)} + \dots, \quad (2.6)$$

$$\frac{\partial}{\partial t} = \epsilon^2 \frac{\partial}{\partial t_2}, \nabla = \epsilon \nabla_1, R = \epsilon^2 R^{(2)}, \quad (2.7)$$

where ϵ is a small parameter. As the Taylor series expansion is applied to Eq. (2.1), the evolution equation can be written as

$$D_i f_i + \frac{\Delta t}{2} D_i^2 f_i = -\frac{1}{\tau \Delta t} (f_i - f_i^{(\text{eq})}) + \bar{\omega}_i R D, \quad (2.8)$$

where $D_i = (\frac{\partial}{\partial t} + c_i \cdot \nabla)$. Substituting Eqs. (2.6) and (2.7) into Eq. (2.8), then,

$$\begin{aligned} & \left(\epsilon^2 \frac{\partial}{\partial t_2} + \epsilon c_i \cdot \nabla_1 \right) (f_i^{(\text{eq})} + \epsilon f_i^{(1)} + \dots) + \left(\epsilon^2 \frac{\partial}{\partial t_2} + \epsilon c_i \cdot \nabla_1 \right)^2 (f_i^{(\text{eq})} + \epsilon f_i^{(1)} + \dots) \\ &= -\frac{1}{\tau \Delta t} (\epsilon f_i^{(1)} + \epsilon^2 f_i^{(2)} + \dots) + \epsilon^2 \bar{\omega}_i R^{(2)} D. \end{aligned} \quad (2.9)$$

One can separate Eq. (2.9) into two parts,

$$c_i \cdot \nabla_1 f_i^{(\text{eq})} = -\frac{1}{\tau \Delta t} f_i^{(1)}, \quad (2.10)$$

$$\frac{\partial f_i^{(\text{eq})}}{\partial t_2} + c_i \cdot \nabla_1 f_i^{(1)} + \frac{\Delta t}{2} (c_i \cdot \nabla_1)^2 f_i^{(\text{eq})} = -\frac{1}{\tau \Delta t} f_i^{(2)} + \bar{\omega}_i R^{(2)} D. \quad (2.11)$$

With the help of Eq. (2.10), Eq. (2.11) is rewritten as

$$\frac{\partial f_i^{(\text{eq})}}{\partial t_2} + \left(\frac{1}{2} - \tau \right) \Delta t (c_i \cdot \nabla_1)^2 f_i^{(\text{eq})} = -\frac{1}{\tau \Delta t} f_i^{(2)} + \bar{\omega}_i R^{(2)} D. \quad (2.12)$$

Since $\sum_i f_i^{(k)} = 0$ ($k \geq 1$), after summing of the Eq. (2.12), so that the PDE is recovered as

$$D \nabla_1^2 u = R^{(2)} D, \quad (2.13)$$

with the aid of the Eq. (2.7), the PDE is rewritten as

$$D \Delta u = D R,$$

which reduces to the Poisson equation if $D \neq 0$.

3. Numerical simulation

Numerical experiments on d -dimension ($d = 1, 2$ and 3) Poisson equations are carried out to validate the model proposed in the present work, and simultaneously, the differences between the present model and previous models are also discussed through a counterexample.

3.1. Numerical examples

Example 1. One-dimension case: The Poisson–Boltzmann equation is first considered, it plays an important role in electrokinetic microchannels [2–4]. The dimensionless form of the equation is

$$\Delta u = \beta \sin(\gamma u), \quad (3.1)$$

where $\gamma = ez\zeta/k_b T$, $\beta = (Kh)^2/\gamma$, $K = \sqrt{\frac{2n_\infty z^2 e^2}{\epsilon \epsilon_0 k_b T}}$ is defined as the reciprocal of the Debye length, $h = 0.8 \mu\text{m}$ is the characteristic length, $n_\infty = c_\infty N_A$ is the bulk ionic number concentration, where $c_\infty = 0.1 \text{ M}$, $N_A = 6.02 \times 10^{23}$. z is the valence of the ions and $|z| = 1$, $e = 1.6201 \times 10^{-19} \text{ C}$ is the absolute value of one proton charge, $k_b = 1.3805 \times 10^{-23} \text{ J mol}^{-1} \text{ K}^{-1}$ is the Boltzmann constant, and $T = 273 \text{ K}$ is the absolute temperature, the dielectric constant $\epsilon \epsilon_0 = 6.95 \times 10^{-10} \text{ C}^2/\text{Jm}$, $\zeta = -10 \text{ mV}$ is the zeta potential. If γu is small enough, $\sin(\gamma u) \approx \gamma u$, which is known as the Debye–Hückel approximation [4]. With the aid of this approximation, Eq. (3.1) can be simplified as

$$\Delta u = k^2 u, \quad (3.2)$$

where $k = \sqrt{\gamma \beta} = Kh \approx 27.79$. Eq. (3.2) has an analytical solution for a specified set of boundary conditions. As an example, the boundary conditions on computational domain $[0, 1]$ are defined as

$$u(0) = 1.0, u(1) = 1.0 \quad (3.3)$$

and analytical solution of the problem is

$$u(x) = \frac{e^k - 1}{e^k - e^{-k}} e^{-kx} + \frac{1 - e^{-k}}{e^k - e^{-k}} e^{kx}. \quad (3.4)$$

We would like to point out that the distribution of u is dependent on k . A parabolic profile is derived if k is small, but a ‘plug’ profile is obtained as k is large enough [3,4].

Example 2. Two-dimension case: The homogeneous Helmholtz equation presented in Ref. [26] is investigated, it can be written as

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - \lambda^2 u = 0 \quad (3.5)$$

with the boundary conditions

$$\begin{aligned} u(0, y) &= \frac{\sinh \mu(1 - y)}{\sinh \mu}, \quad u(1, y) = -\frac{\sinh \mu(1 - y)}{\sinh \mu}, \\ u(x, 0) &= \cos \pi x, \quad u(x, 1) = 0, \end{aligned} \quad (3.6)$$

where $\mu = \sqrt{\lambda^2 + \pi^2}$, and λ is set to be 2 in the following numerical simulation. Analytical solution of this problem is

$$u(x, y) = \cos \pi x \frac{\sinh \mu(1 - y)}{\sinh \mu}.$$

Example 3. Three-dimension case: The last example is implemented on three-dimension space. The Poisson equation is defined on $[0, 1] \times [0, 1] \times [0, 1]$, and written as

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \sin(u^2) + f. \quad (3.7)$$

The boundary conditions are defined as

$$\begin{aligned} u(0, y, z) &= \sin(y + z), \quad u(x, 0, z) = \sin(x + z), \quad u(x, y, 0) = \sin(x + y), \\ u(1, y, z) &= \sin(1.0 + y + z), \quad u(x, 1, z) = \sin(1.0 + x + z), \quad u(x, y, 1) = \sin(1.0 + x + y), \end{aligned} \quad (3.8)$$

where $f = -\sin(\sin^2(x+y+z)) - 3\sin(x+y+z)$. The analytical solution to this problem is $u = \sin(x+y+z)$.

3.2. Numerical results

Numerical simulations are carried out on a 100, 100×100 and $100 \times 100 \times 100$ lattice for three different cases. c is set to be 1.0, the dimensionless relaxation time $\tau = 1.0$, the distribution function $f_i(x, t)$ is initialized by setting to equal $f_i^{(eq)}(x, t)$, and u is set to be 0 at $t = 0$ except on the boundary. The non-equilibrium extrapolation scheme proposed by Guo et al. [27] is used for boundary conditions. In addition, the global relative error (GRE) is used to test the accuracy of the proposed model, and is defined by

$$\text{GRE} = \frac{\sum |u - u^*|}{\sum |u|},$$

where u , u^* are analytical and numerical solutions.

A simple model D1Q3 is chosen to solve the Poisson–Boltzmann equation (3.1) with the boundary conditions (3.3). The possible velocity direction e_i in D1Q3 model are $e_0 = 0$, $e_1 = 1$, $e_2 = -1$. $\omega_0 = 2/3$, $\omega_i = 1/6$ ($i = 1, 2$), $\alpha = 1/3$, one can derive the value of α from Eq. (2.3), $\bar{\omega}_0 = 0$, $\bar{\omega}_i = 1/2$ ($i = 1, 2$). Stability analysis [28] on incompressible model proposed by Guo et al. [24] can also be used on the present model. Detailed numerical results are shown in Fig. 1 and Table 1, and GRE = 0.0113 for this example. On the other hand, as shown in Fig. 2, the proposed model is also a second order scheme on precision (the slope of the line is about 1.97), which is the same as conventional models.

Subsequently, D2Q5 model is applied to solve homogeneous Helmholtz equation (3.5) with the boundary conditions (3.6). e_i in this model are $e_0 = (0, 0)$ and $e_i = (\cos[(i-1)\pi/2], \sin[(i-1)\pi/2])$ ($i = 1, 2, 3, 4$). $\omega_0 = 0$, $\omega_i = 1/4$ ($i = 1-4$), $\bar{\omega}_0 = 0$, $\bar{\omega}_i = 1/4$ ($i = 1-4$), and $\alpha = 1/2$. Numerical results and analytical results can be found in Fig. 3 and Table 2. The global relative error GRE = 2.3180×10^{-5} .

Finally, D3Q13 model is used to solve the Poisson equation (3.7) with the boundary conditions (3.8). The possible velocity direction e_i and some parameters in this model are set as follows. $e_0 = (0, 0, 0)$, $e_1 = (1, 1, 0)$, $e_2 = (1, -1, 0)$, $e_3 = (1, 0, 1)$, $e_4 = (1, 0, -1)$, $e_5 = (0, 1, 1)$, $e_6 = (0, 1, -1)$, $e_7 = (-1, -1, 0)$, $e_8 = (-1, 1, 0)$, $e_9 = (-1, 0, 1)$, $e_{10} = (-1, 0, -1)$, $e_{11} = (0, -1, -1)$, $e_{12} = (0, -1, 1)$, $\omega_0 = 1/2$, $\omega_i = 1/24$ ($i = 1-12$), $\bar{\omega}_0 = 0$, $\bar{\omega}_i = \frac{1}{12}$ ($i = 1-12$), and $\alpha = 1/3$. Numerical and analytical results are shown in Fig. 4 and Table 3, GRE = 2.3836×10^{-5} for this case.

All numerical results listed above show that the Poisson equations can be solved efficiently and accurately with the new lattice Boltzmann model.

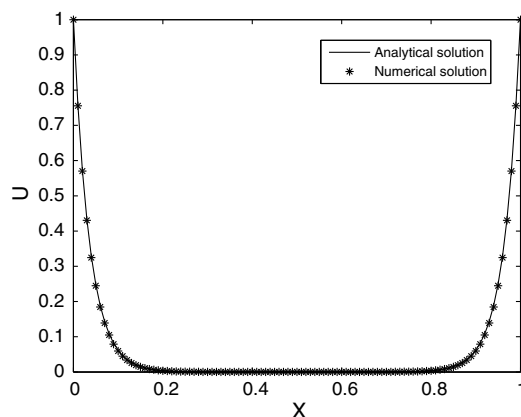


Fig. 1. Approximated analytical solution (Eq. (3.4)) and numerical solution of Eq. (3.1) with boundary conditions (3.3).

Table 1
Comparisons between numerical solution and approximated analytical solution at some grids

x_i	Numerical solution	Analytical solution	Absolute error: $ u(x_i) - u^*(x_i) $
0	1.000000	1.000000	0.000000
0.1	0.059628	0.062100	0.002472
0.2	0.003538	0.003900	0.000362
0.3	0.000210	0.000200	1.0e–005
0.4	1.2e–005	0.000000	1.2e–005
0.5	1.0e–006	0.000000	1.0e–006
0.6	1.2e–005	0.000000	1.2e–005
0.7	0.000210	0.000200	1.0e–005
0.8	0.003538	0.003900	0.000362
0.9	0.059628	0.062100	0.002472
1.0	1.000000	1.000000	0.000000

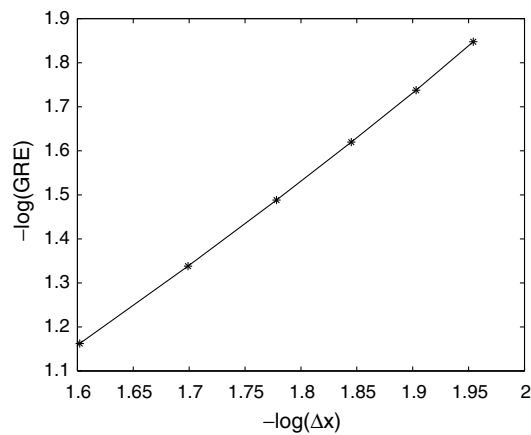
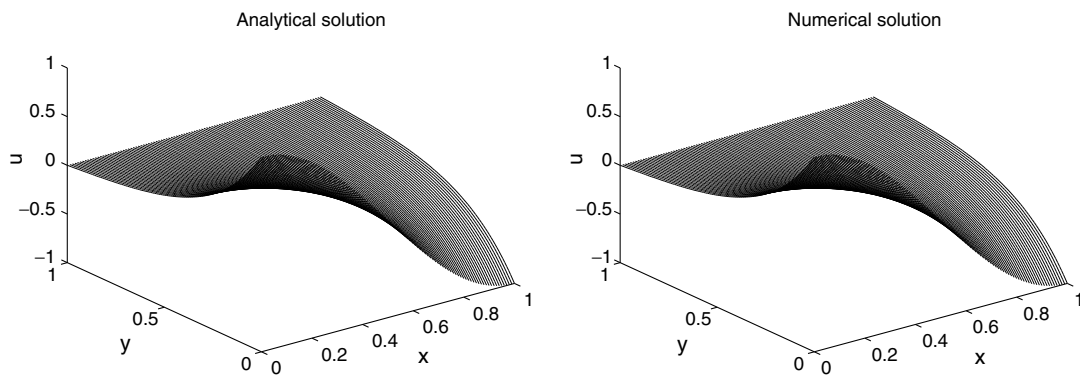


Fig. 2. The relation between GRE and lattice spacing.



(a) Analytical solution of Eq. (3.5) with boundary conditions (3.6).

(b) Numerical solution of Eq. (3.5) with boundary conditions (3.6).

Fig. 3. (a) Analytical and (b) numerical solution of Eq. (3.5) with boundary conditions (3.6).

Table 2
Comparisons between numerical solution and analytical solution at some grids

(x_i, y_i)	Numerical solution	Analytical solution	Absolute error: $ u(x_i, y_i) - u^*(x_i, y_i) $
(0, 0)	1.00000000	1.00000000	0.00000000
(0.1, 0.1)	0.65492600	0.65491768	0.00000832
(0.2, 0.2)	0.38337310	0.38336116	0.00001194
(0.3, 0.3)	0.19138370	0.19137431	0.00000939
(0.4, 0.4)	0.06891276	0.06890836	0.00000440
(0.5, 0.5)	0.00000000	0.00000000	0.00000000
(0.6, 0.6)	−0.03141794	−0.03141549	0.00000245
(0.7, 0.7)	−0.03873976	−0.03873687	0.00000289
(0.8, 0.8)	−0.03186931	−0.03186734	0.00000197
(0.9, 0.9)	−0.01750391	−0.01750324	0.00000067
(1.0, 1.0)	0.00000000	0.00000000	0.00000000

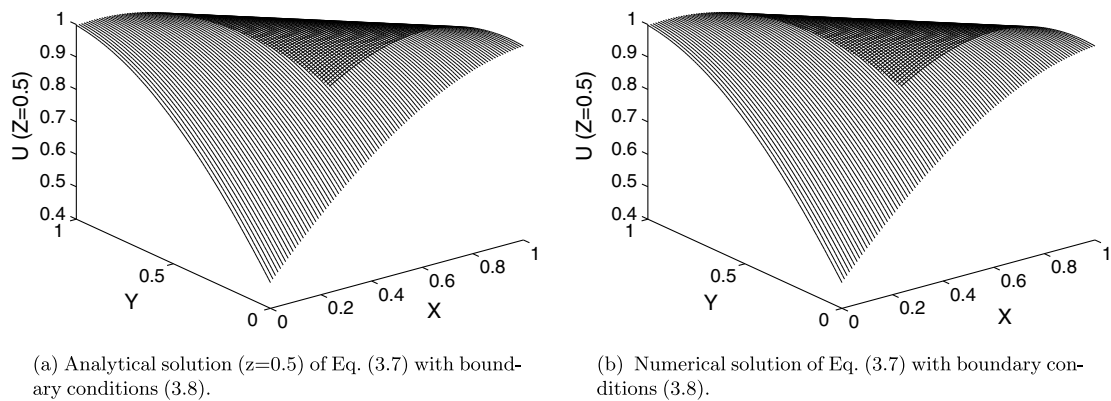


Fig. 4. (a) Analytical and (b) numerical solutions ($z = 0.5$) of Eq. (3.5) with boundary conditions (3.6).

Table 3
Comparisons between numerical solution and analytical solution at some grids

(x_i, y_i, z_i)	Numerical solution	Analytical solution	Absolute error: $ u(x_i, y_i, z_i) - u^*(x_i, y_i, z_i) $
(0, 0, 0)	0.000000	0.000000	0.000000
(0.1, 0.1, 0.1)	0.295518	0.295520	0.000002
(0.2, 0.2, 0.2)	0.564626	0.564642	0.000016
(0.3, 0.3, 0.3)	0.783285	0.783327	0.000042
(0.4, 0.4, 0.4)	0.931972	0.932039	0.000067
(0.5, 0.5, 0.5)	0.997417	0.997495	0.000078
(0.6, 0.6, 0.6)	0.973780	0.973848	0.000068
(0.7, 0.7, 0.7)	0.863167	0.863209	0.000042
(0.8, 0.8, 0.8)	0.675446	0.675463	0.000017
(0.9, 0.9, 0.9)	0.427377	0.427380	0.000003
(1.0, 1.0, 1.0)	0.141120	0.141120	0.000000

3.3. Counterexample

In this subsection, a counterexample is presented to demonstrate the difference between the proposed model and some artificial models [1–3,23]. For simplicity, but without loss of generality, the counterexample is only presented for one-dimension case.

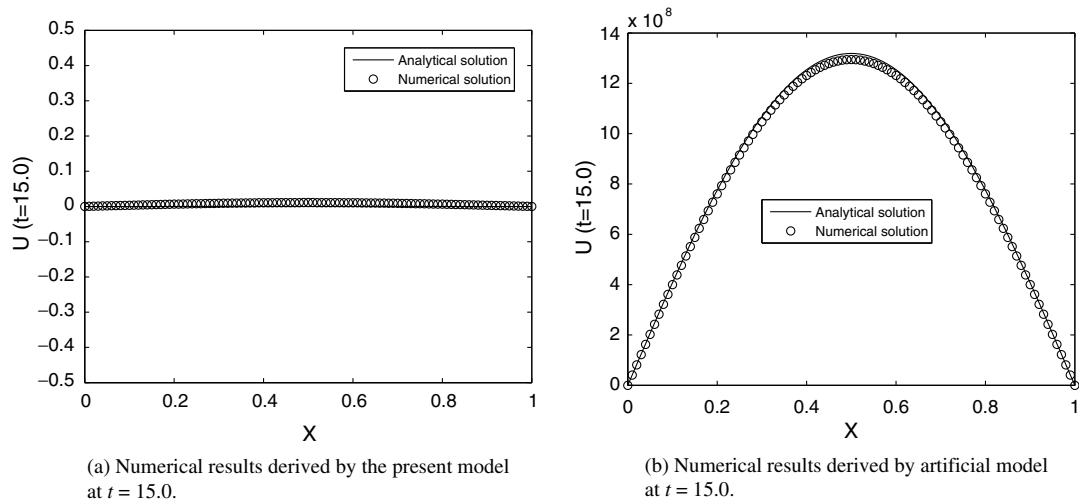


Fig. 5. Numerical results for comparison between the present model and artificial model.

The Poisson equation is defined as

$$\frac{2\Delta u}{\pi^2} + 9u = 0 \quad (3.9)$$

with the following boundary conditions

$$u(0) = u(1) = 0 \quad (3.10)$$

there is a constant solution $u(x) = 0$ for this problem.

The present model and artificial model are applied to solve Eq. (3.9) with boundary conditions (3.10), and the same initialization $u(x) = \sin(\pi x)$ is used in these two models. It is clear that the numerical solution derived by the present model converges to 0, i.e., analytical solution of the Poisson equation (3.9) (see Fig. 5a). However, as shown in Fig. 5b, the numerical solution derived by artificial model converges to $u(x, t) = e^{7t/5} \sin(\pi x)$, which is the analytical solution of the diffusion equation

$$\frac{\partial u}{\partial t} = \frac{2\Delta u}{5\pi^2} + \frac{9u}{5}. \quad (3.11)$$

The reason for this great difference is that, the equation solved by artificial model is the diffusion equation (3.11), not the Poisson equation (3.9).

To test the effect of the initialization on the numerical results, the initialization $u = 0$ has also been used in the present model and the artificial model. It is found that the numerical solutions of the two models converge to analytical solution of the Poisson equation ($u = 0$).

It is evident that the artificial model is seriously dependent on initialization, and some unexpected errors may be induced by initialization when the artificial model is used to solve the Poisson equation. In fact, the same problem exists in all artificial models [1–3,23].

4. Summary and conclusion

In the present work, a novel lattice Boltzmann model is proposed to solve d -dimension Poisson equation. The efficiency and accuracy of the proposed model are validated through detailed numerical simulations. In addition, some differences between present model with the artificial models should be noted: (1) Unlike existing artificial models, which can be viewed as the solvers to the diffusion equation, the present model directly aims to solve the Poisson equation. (2) Due to the fact that the artificial models are solvers to diffusion equation, the numerical results derived by artificial models are dependent on initialization, i.e., different

initializations may lead to different numerical solutions. However, the model proposed in this work is independent of initialization. (3) The approach to compute physical quantity u is different from existing lattice Boltzmann models, and in practical numerical simulations, only $(q - 1)$ -velocity is used in the present $DdQq$ model, so that the computation expense can be decreased to some extent.

Acknowledgements

We would like to thank anonymous referees for their valuable comments which greatly improved the work. One of authors (Z. Chai) would like to thank Prof. Z. Guo and Dr. L. Zheng for their useful discussions during this work. This study is financially supported by the National Basic Research Program of China (Grant No. 2006CB705800) and the National Natural Science Foundation of China (Grant No. 50606012).

References

- [1] X. He, N. Ling, Lattice Boltzmann simulation of electrochemical systems, *Comput. Phys. Commun.* 129 (2000) 158–166.
- [2] Z. Guo, T.S. Zhao, Y. Shi, A lattice Boltzmann algorithm for electro-osmotic flows in microfluidic devices, *J. Chem. Phys.* 122 (2005) 144907.
- [3] J. Wang, M. Wang, Z. Li, Lattice Poisson–Boltzmann simulations of electro-osmotic flows in microchannels, *J. Colloid. Interface Sci.* 296 (2006) 729–736.
- [4] Z. Chai, B. Shi, Simulation of electro-osmotic flow in microchannel with lattice Boltzmann method, *Phys. Lett. A* 364 (2007) 183–188.
- [5] H.I. Siyyam, An accurate solution of the Poisson equation by the finite difference–Chebyshev–Tau method, *Appl. Math. Mech.* 22 (8) (2001) 0935–0939.
- [6] A.H. Boschitsch, M.O. Fenley, Hybrid boundary element and finite difference method for Solving the Nonlinear Poisson–Boltzmann equation, *J. Comput. Chem.* 25 (7) (2004) 935–955.
- [7] Z. Cai, S. Kim, S. Kim, S. Kong, A finite element method using singular functions for Poisson equations: mixed boundary conditions, *Comput. Methods Appl. Mech. Engrg.* 195 (2006) 2635–2648.
- [8] F. Kikuchi, H. Saito, Remarks on a posteriori error estimation for finite element solutions, *J. Comput. Appl. Math.* 199 (2) (2007) 329–336.
- [9] S.J. Liao, General boundary element method for Poisson equation with spatially varying conductivity, *Eng. Anal. Bound. Elem.* 21 (1998) 23–38.
- [10] R. Benzi, S. Succi, M. Vergassola, The lattice Boltzmann equation: theory and applications, *Phys. Rep.* 222 (1992) 147–197.
- [11] S. Chen, G.D. Doolen, Lattice Boltzmann method for fluid flows, *Annu. Rev. Fluid Mech.* 30 (1998) 329–364.
- [12] H. Chen, S. Kandasamy, S. Orszag, R. Shock, S. Succi, V. Yakhot, Extended Boltzmann kinetic equation for turbulent flows, *Science* 301 (2003) 633–636.
- [13] U. Frisch, B. Hasslacher, Y. Pomeau, Lattice-gas automata for the Navier–Stokes equation, *Phys. Rev. Lett.* 56 (1986) 1505–1508.
- [14] X. He, L.S. Luo, A priori derivation of the lattice Boltzmann equation, *Phys. Rev. E* 55 (1997) R6333–R6336.
- [15] X. Shan, X. He, Discretization of the velocity space in the solution of Boltzmann equation, *Phys. Rev. Lett.* 80 (1998) 65–68.
- [16] Z.L. Guo, B.C. Shi, N.C. Wang, Fully Lagrangian and lattice Boltzmann methods for the advection–diffusion equation, *J. Sci. Comput.* 14 (3) (1999) 291–300.
- [17] B. Deng, B.C. Shi, G.C. Wang, A new lattice Bhatnagar–Gross–Krook model for convection–diffusion equation with a source term, *Chin. Phys. Lett.* 22 (2) (2005) 267–270.
- [18] S. Chen, S.P. Dawson, G.D. Doolen, D.R. Janecky, A. Lawniczak, Lattice methods and their applications to reacting systems, *Comput. Chem. Engrg.* 19 (6–7) (1995) 617–646.
- [19] C.Y. Zhang, H.L. Tan, M.R. Liu, L.J. Kong, A lattice Boltzmann model and simulation of KdV–Burgers equation, *Commun. Theor. Phys.* 42 (2) (2004) 281–284.
- [20] C. Ma, A new lattice Boltzmann model for KdV–Burgers equation, *Chin. Phys. Lett.* 29 (9) (2005) 2313–2315.
- [21] Z. Chai, B. Shi, L. Zheng, A unified lattice Boltzmann model for some nonlinear partial differential equations, *Chaos Solitons Fractals*, in press, doi:10.1016/j.chaos.2006.07.023.
- [22] G. Yan, A lattice Boltzmann equation for waves, *J. Comput. Phys.* 161 (2000) 61–69.
- [23] M. Hirabayashi, Y. Chen, H. Ohashi, The lattice BGK model for the Poisson equation, *JSME Int. J. Ser. B* 44 (1) (2001) 45–52.
- [24] Z. Guo, B. Shi, N. Wang, Lattice BGK model for incompressible Navier–Stokes equation, *J. Comput. Phys.* 165 (2000) 288–306.
- [25] S.P. Dawson, S. Chen, D.G. Doolen, Lattice Boltzmann Computations for reaction–diffusion equations, *J. Chem. Phys.* 98 (2) (1993) 1514–1523.
- [26] A. Averbuch, L. Vozovoi, M. Israeli, On a fast direct elliptic solver by a modified Fourier method, *Numer. Algorithms* 15 (1997) 287–313.
- [27] Z.L. Guo, C.G. Zheng, B.C. Shi, Non-equilibrium extrapolation method for velocity and pressure boundary conditions in the lattice Boltzmann method, *Chin. Phys.* 11 (4) (2002) 366–374.
- [28] M.K. Banda, W.A. Yong, A. Klar, A stability notion for lattice Boltzmann equations, *SIAM J. Sci. Comput.* 27 (2006) 2098–2111.